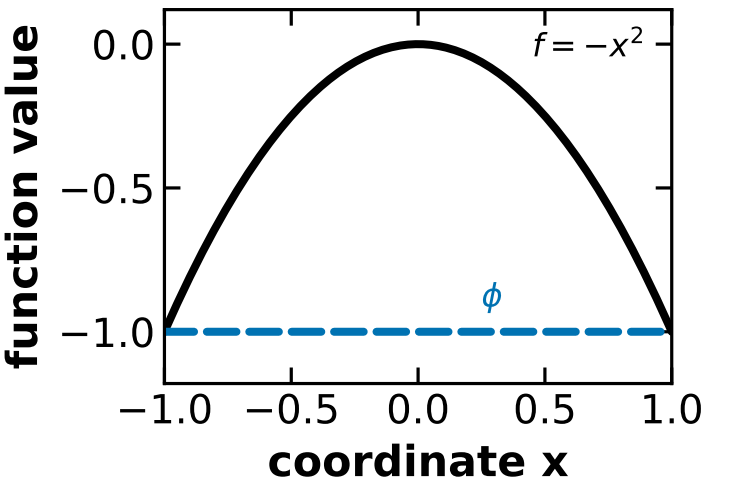

$$\phi(x) = f(x) - \alpha(b - x)(x - a)$$

$$\phi'' = f'' + 2\alpha$$

Here a = -1, b = 1, alpha = 1.

Curvature must be valid throughout the box.